

Release Notes for

AT32F423 Firmware Library

V2.1.0-2026/06/02

1. CMSIS
 - system_at32f423.c
 - ◆ Updated the process of the System_init() to avoid potential clock failure during switching
2. Middlewares
 - I2C
 - ◆ Corrected these macro definition typos:
I2C_MEM_ADDR_WIDIH_8 is changed to I2C_MEM_ADDR_WIDTH_8;
I2C_MEM_ADDR_WIDTH_8 is changed to I2C_MEM_ADDR_WIDTH_16
 - USB
 - ◆ Updated the usbh_enum_handler() in host mode by adding a delay of 10ms after the SET_ADDRESS operation to ensure that the device address is taken into account before starting subsequent operation and configuration
 - ◆ Updated the msc class by adding a step to handle the test_unit() and the start_stop() to enhance compatibility
 - ◆ Updated the msc class to address the problem of compiling warning issued in the uhost_process_handler() when the USBH_DEBUG_ENABLE is not activated
3. Drivers
 - PWC
 - ◆ __force_stores() is replaced with the __ISB() and __DSB() to adapt to different compilers
 - CRM
 - ◆ Updated the processes of the crm_reset(), crm_hick_sclk_frequency_select() and the crm_hick_divider_select() to avoid potential clock failure during switching
 - EXINT
 - ◆ Added a new EXINT_TRIGGER_NONE_EDGE to adapt to software trigger without edge configuration requirement
4. Demos
 - I2C
 - ◆ Corrected these macro definition typos:
I2C_MEM_ADDR_WIDIH_8 is changed to I2C_MEM_ADDR_WIDTH_8;
I2C_MEM_ADDR_WIDIH_16 is changed to I2C_MEM_ADDR_WIDTH_16
5. Updated some notes

V2.0.9-2025/11/24

1. CMSIS
 - at32f423.h
 - ◆ Removed unnecessary base address macros
2. Drivers

- ADC
 - ◆ Removed unnecessary bit name “spt_bit.cspt18” and the register `csts_bit`
 - CRM
 - ◆ Changed the bit name from `adc14rst` to `tmr14rst`
 - ◆ Removed excess definitions in the `CRM_xxx_PERIPH_LOWPOWER`
 - TMR
 - ◆ Updated these functions: `tmr_pwm_input_config()`, `tmr_encoder_mode_config()` and `tmr_output_channel_config()`
The channel must be disabled before configuring channel mode, otherwise it fails to switch between channel modes
3. Middlewares
- USB
 - ◆ Updated “keyboard class” to add a new member “send_state” in the `keyboard_type` structure to record send state, and updated the flow control handling in the `usb_keyboard_class_send_report`, so as to avoid data error
 - ◆ Updated “audio class” to change the audio sampling frequency request configuration, so as to solve host compatibility issue
4. Demos
- ADC
 - ◆ Updated ADC initialization process in all demos to avoid errors occurring under extreme circumstances
 - ◆ Updated DMA-related demos to enable DMA circular mode during configuration to achieve circular trigger
 - ◆ Removed the `software_trigger_repeat` demo
 - USB
 - ◆ Updated the clock division configuration process in the `usb_device` demo to prevent USB clock division configuration error as a result of AHB division
 - ◆ Adjusted the procedure of configuring Deepsleep extra low-power mode in keyboard, and updated the latency method to achieve stabilization after wakeup
 - ◆ Adjusted the procedure of configuring Deepsleep extra low-power mode in mouse, and updated the latency method to achieve stabilization after wakeup
 - ◆ Updated the `USB_FLASH_ADDR_OFFSET` value in `msc` from `0x08005000` to `0x08008000`, so as to solve compiling range overflow issue in different environments
 - ◆ Updated Jump APP function in the `virtual_msc_iap` demo to add compiling non-optimization settings, so as to avoid Jump failure arising from optimization settings
 - templates
 - ◆ Updated the “include” path in the `eclipse_gcc` to prevent compiling error due to incorrect parameters in the `workspace_project_locations` during multi-project import process
 - Board
 - ◆ Updated the description of some parameter variables in `board.c` to solve warning issued when using a version of compiler
 - Cortex-m4

- ♦ Updated the description of latency judgement in the `systick_interrupt` from “greater than” to “greater than or equal to”, to avoid one more 1ms than it should have taken to toggle IO
 - AT32IDE and Eclipse projects
 - ♦ Removed “release” items and reserved “debug” items
 - IAP
 - ♦ Updated all IAP_Programmer.exe
 - Keil projects
 - ♦ Updated the IROM size configuration in some Keil projects
5. Updated some notes

V2.0.8-2025/01/20

6. Drivers
- PWC
 - ♦ Updated LDO output level settings, removed LDO configuration option “PWC_LDO_OUTPUT_1V0” and added “PWC_LDO_OUTPUT_1V1”.
7. Middlewares
- USB
 - ♦ Updated “`usbh_hch_in_handler()`” function in host mode, and channels are disabled during interrupt transfer and synchronous transfer when NAK occurs.
8. Demos
- IAP
 - ♦ Updated IAP_Programmer.exe.
 - USB_DEVICE
 - ♦ Updated USB low-power wakeup configuration and process to enhance system stability when entering/existing low-power mode.
 - SPI
 - ♦ Updated the structure and framework of all SPI demo codes, removed “`usb_jtagpin_hardwarecs_dma`” and added “`fullduplex_dma`”.
9. Updated some notes.

V2.0.7-2024/12/13

1. CMSIS
- DSP
 - ♦ Updated DSP source code library.
2. Drivers
- CAN
 - ♦ Updated “`can_transmit_cancel()`” function writing to avoid cross-mailbox interference due to misoperation.
 - USB。
 - ♦ Updated “`usb_hch_halt()`” function in host mode and improved channel halt processing flow.
 - ♦ Updated “`usb_hc_enable()`” function in host mode to ensure that its original

state is cleared when enabling the corresponding channel.

- ♦ Updated “usb_host_disable()” function in host mode, and updated process of disabling channels to avoid inadvertently opening the remaining channels.
- ♦ Updated “usb_ept_open()” function in device mode, and added synchronous transfer handling (frame number ignored).
- ♦ Updated “usb_ept_close()” function in device mode, and improved endpoint closing process.

3. Middlewares

- USB

- ♦ Updated “usbd_reset_handler()” function, and changed endpoint status to NAK on USB RESET.
- ♦ Added “usbd_ept_in_check_fifo()” function to check the current endpoint status and FIFO when transmitting data through USB.
- ♦ Updated *mouse_class.c* and *custom_hid_class.c*, and added the process of checking whether the previous packet has been transferred completely when sending hid/mouse data.

4. Demos

- I2C

- ♦ Added memory_write demo.
- ♦ Updated eeprom, and added multi-page write logic.

- Utilities

- ♦ Added support for AT32IDE project in SLIB demo.

- PWC

- ♦ Updated the delay mechanism after waking up from DeepSleep mode in all PWC related demos to ensure stability.

5. Updated some notes.

V2.0.6-2024/08/08

1. Middlewares

- USB

- ♦ Revised STALL state assignment method in “usbh_hch_out_handler()” function.
- ♦ Updated “custom hid report” and “audio hid report” descriptors.
- ♦ Replaced keyword “__weak” with “__WEAK” to improve cross-platform compatibility.

2. Demos

- Cortex-m4

- ♦ Modified IROM base address and size settings of Keil project in cmsis_dsp.
- ♦ Modified code initialization sequence and fixed AC5 compilation problems.

- I2C

- ♦ Modified “GPIO_PULL_UP” to “GPIO_PULL_NONE” in I2C GPIO initialization to enhance the adaptability of external circuits.
- ♦ Modified DMA channel configuration sequence in “communication_dma” to prevent illegal pointer access.

- PWC
 - ◆ Added “pwc_voltage_regulate_set (PWC_REGULATOR_EXTRA_LOW_POWER);” in all PWC demos to achieve optimal power consumption performance in low-power mode.
 - USB
 - ◆ Modified write handling of *flash_fat16.c* file in *virtual_msc_iap* to enhance compatibility.
 - Utilities
 - ◆ Adjusted non-optimization option “__attribute__” in “*usart_iap_demo*” and “*usb_iap_demo*” to comply with AC6 compiler.
 - ◆ Modified RAM size in IAR project link file of *slib_demo/project_l1*.
3. Updated some notes.

V2.0.5-2024/01/26

1. Middlewares
 - USB
 - ◆ Updated the type of “*usbd_core_type*” structure members to ensure 4-byte-alignment of “*default_config*” and “*dev_config*” addresses.
 - ◆ Updated “custom hid report” and “audio hid report” descriptors.
2. Updated some notes.

V2.0.4-2024/01/05

1. Updated timer input_capture demo counting method.
2. Fixed HID low-speed identification problems in USB demo.
3. Updated systick initialization function in systick interrupt demo.
4. Added winusb demo.
5. Added ADC loop_transfer demo.
6. Updated the method to call “*xx_interrupt_flag_get()*” function in all demos.

V2.0.3-2023/10/26

1. Updated some notes and readme.
2. Updated drivers, see drivers release notes for details.

V2.0.2-2023/08/04

1. Added AT32 IDE project to demos under utilities.
2. Added CRC polynomial support.
3. Updated USB driver and code process.
4. Improved redirecting compatibility while using printf in IAR9.
5. Added AT32F423CBU7-7 as a new part number.
6. Added CAN demo.
7. Updated some notes.

V2.0.1-2023/04/26

1. Updated USB MSC files and demo to support 64-bit address.

2. Revised buffer 4 byte alignment issue in USB demo.
3. Revised the value of USD_BASE macro definition.
4. Added macro definitions regarding Flash size and Flash demo.
5. Updated some notes.

V2.0.0-2023/03/24

1. Initial release of AT32F423 firmware library.